



Management Science

Publication details, including instructions for authors and subscription information:
<http://pubsonline.informs.org>

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To cite this article:

Saif Benjaafar ; , Zicheng Wang ; , Xiaotang Yang (2025) Human in the Loop Automation: Ride-Hailing with Remote (Tele-)Drivers. Management Science 71(3):2527-2543. <https://doi.org/10.1287/mnsc.2022.01687>

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Human in the Loop Automation: Ride-Hailing with Remote (Tele-)Drivers

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Received: June 8, 2022

Revised: May 31, 2023; November 14, 2023

Accepted: January 6, 2024

Published Online in Articles in Advance:
June 10, 2024

<https://doi.org/10.1287/mnsc.2022.01687>

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Abstract. Tele-driving refers to a novel concept by which drivers can remotely operate vehicles (without being physically in the vehicle). By putting the human back in the loop, tele-driving has emerged recently as a more viable alternative to fully automated vehicles with ride-hailing (and other on-demand transportation-enabled services) being an important application. Because remote drivers can be operated as a shared resource (any driver can be assigned to any customer regardless of trip origin or destination), it may be possible for such services to deploy fewer drivers than vehicles without significantly reducing service quality. In this paper, we examine the extent to which this is possible. Using a spatial queueing model that captures the dynamics of both pickup and trip times, we show that the impact of reducing the number of drivers depends crucially on system workload relative to the number of vehicles. In particular, when workload is sufficiently high relative to the number of vehicles, we show that, perhaps surprisingly, reducing the number of drivers relative to the number of vehicles can actually improve service level (e.g., as measured by the amount of demand fulfilled in the case of impatient customers). Having fewer drivers than vehicles ensures that there are always idle vehicles; the fewer the drivers, the likelier it is for there to be more idle vehicles. Consequently, the fewer the drivers, the likelier it is for the pickup times to be shorter (making overall shorter service times likelier). The impact of shorter service time is particularly significant when the workload is high, and in this case, it is enough to overcome the loss in driver capacity. When workload is sufficiently low relative to the number of vehicles, we show that it is possible to significantly reduce the number of drivers without significantly reducing service level. In systems in which customers are patient and willing to queue up for the service, we show that reducing the number of drivers can also reduce delay, including stabilizing a system that may otherwise be unstable. In general, relative to a system in which the number of vehicles equals the number of drivers (as in a system with in-vehicle drivers), a system with remote drivers can result in savings in the number of drivers either without significantly degrading performance or actually improving performance. We discuss how these results can, in part, be explained by the interplay of two counteracting forces: (1) having fewer drivers increasing service rate and (2) having fewer drivers reducing the number of servers with the relative strength of these forces depending on system workload.

History: Accepted by Baris Ata, stochastic models and simulation.

Funding: This work was supported by the US National Science Foundation [Grant SCC-1831140], and the Guangdong (China) Provincial Key Laboratory of Mathematical Foundations for Artificial Intelligence [2023B1212010001].

Supplemental Material: The online appendix and data files are available at <https://doi.org/10.1287/mnsc.2022.01687>.

Keywords: tele-driving • ride hailing • spatial queueing systems • capacity optimization

1. Introduction

It is becoming increasingly apparent that fully autonomous vehicles (AVs) may take longer than initially expected to become a reality. There is growing consensus that large-scale deployment of level 5 autonomy (full autonomy) is unlikely to happen soon (Doll et al.

2020). Even if the technological challenges were to be surmounted in the near future, the public acceptance of full autonomy may take longer. In particular, concerns about safety continue to be high along with unease about delegating to machines critical life-and-death decisions. A new technology that combines vehicles

that are nearly autonomous with tele-operators has recently emerged with the potential of overcoming both the technological hurdles and the societal concerns of driverless vehicles. Tele-operated vehicles may provide the efficiency and flexibility of autonomous vehicles, keeping humans in the loop and ultimately responsible for driving decisions.

Several pilots are demonstrating the commercial viability of tele-driving. Vay, a German car-sharing company, has begun testing a car-sharing service in which vehicles are operated by tele-drivers from a central command center (Blanco 2021). Vay's stated aim is to offer an Uber-like service with remote drivers (Blanco 2021). Halo is a U.S.-based company collaborating with T-Mobile, a telecommunication company, to pilot a robo-taxi service with 5G-powered vehicles that can be operated remotely (Pegoraro 2021). The viability of tele-driving has been demonstrated in other settings, including food delivery using remotely controlled robots (e.g., Coco 2022, Starship 2024), material handling in factories and warehouses (e.g., remotely operated forklift trucks; PhantomAuto 2022), and remotely operated trucks (Sawers 2022). In these and other applications, tele-driving is being enabled by advances in wireless communication (e.g., 5G technology and Amazon Web Services wavelength technology to minimize latency), video compression (to enable remote video display), and edge computing (to allow for in-vehicle data processing and controls), among others; see Sawers (2020) for further discussion.

Tele-operated vehicles in the context of ride hailing (the focus of this paper) offer several advantages (over conventional vehicles with in-vehicle drivers). Perhaps the most important of these is the ability to treat remote drivers as a common resource. Whereas vehicles continue to be constrained by their geographic location, remote drivers can be interchangeably assigned to fulfill trips regardless of a trip origin or destination. Moreover, the fact that each driver is not dedicated to a single vehicle implies that a service could operate with fewer drivers than vehicles—important in settings in which drivers are costly or are in short supply.

In this paper, we examine the extent to which this is possible. Specifically, we are interested in investigating the impact of operating with fewer remote drivers than vehicles on system performance; performance measures of interest include the amount of demand that can be fulfilled and the amount of delay customers experience.¹ The setting we consider is of a service that operates over a continuous region. Requests for transport (from an origin to a destination) arise continuously over time. The operator of the service manages a fleet of vehicles and a pool of remote drivers. We consider two settings: one in which customers are impatient and one in which they are patient. In the case of impatient customers, a customer leaves the system without receiving service if the

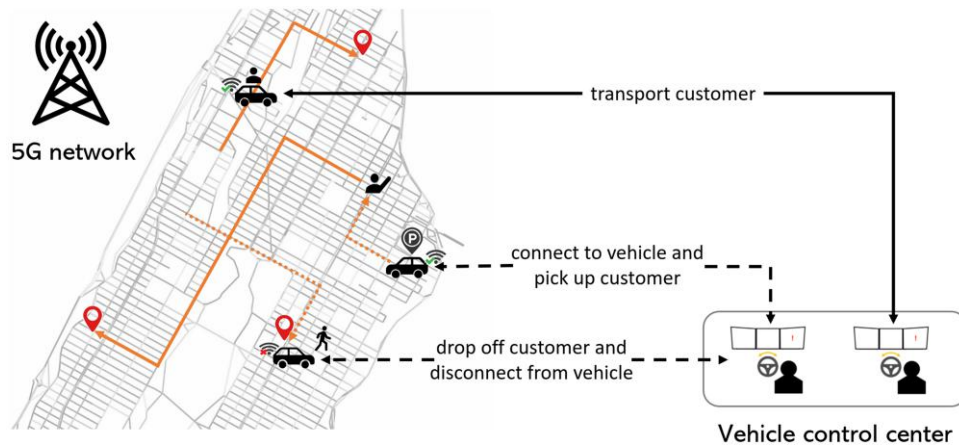
customer cannot be immediately matched with a vehicle and a driver. That is, if there are no idle drivers/vehicles when a customer makes a trip request, the request cannot be fulfilled, and the customer is considered lost. In the case of patient customers, a customer is willing to wait until both a vehicle and a driver are available to serve the customer. The operator of the service decides on which vehicles to match with which customers (remote drivers are assumed to be homogeneous, and any available driver can be matched with any customer–vehicle pair). Once a vehicle–driver pair is assigned to a customer, the driver takes over the control of the vehicle and drives it (remotely) to the location of the customer. Once the customer is picked up, the remote driver drives the vehicle to the customer's requested destination. Upon trip completion, the driver and the vehicle are uncoupled and become available to be independently assigned to future customer requests (see Figure 1 for a graphical illustration).

Note that the setting we consider is one in which the number of vehicles and drivers are determined by the service provider. The service provider is also in control of decisions about matching requests with vehicles and drivers and vehicle repositioning (i.e., we do not allow for drivers to act strategically).

We show how the dynamics of the system can be approximately captured by a multiserver queueing model with state-dependent service time. We use this model to examine the impact of varying the number of drivers using a benchmark case in which the number of drivers equals the number of vehicles. The following is a summary of our main findings. We discuss first the case of impatient customers:

- There are three distinct operating regimes in which the effect of varying the number of remote drivers varies: supply-limited, intermediate, and supply-rich regimes, in which “supply” refers to the number of vehicles relative to workload. In the supply-limited regime, we show that reducing the number of drivers (relative to the number of vehicles) can actually lead to a higher service level as measured by the fraction of demand fulfilled. We provide a necessary and sufficient condition for this to occur and show that the improvement in service level can be significant. In general, in this regime, the effect of varying the number of drivers on service level (all else staying the same) is nonmonotonic.
- In the intermediate regime, we show that reducing the number of remote drivers can similarly lead to a higher service level. We provide a necessary and sufficient condition for this to occur. However, when it occurs, the improvement can be insignificant (asymptotically, the improvement is negligible).
- In the supply-rich regime, we show that, although reducing the number of drivers always leads to a lower service level, the decrease in service level can be small even when the decrease in the number of drivers is

Figure 1. (Color online) An Illustration of a Ride-Hailing Service with Tele-Drivers



significant (asymptotically, this decrease is at least a half). In general, we provide a well-specified bound on the relationship between service level and the number of drivers when the number of drivers is sufficiently large.

The following is a summary of our main findings for systems with patient customers:

- Here, too, there are three distinct regimes. In the supply-limited regime, the system, regardless of the number of drivers, is unstable (customer delay is not finite).
- In the intermediate regime, we show that reducing the number of drivers can make a system that is otherwise unstable stable.
- In the supply-rich regime, we show that a result similar to the one obtained for impatient customers holds. That is, it is possible to significantly decrease the number of drivers and only marginally increase expected customer delay (asymptotically, the number of drivers can be decreased by at least half).

Finally, we provide results from numerical experiments using real-world taxi data from the city of New York.

1.1. An Intuitive Explanation

The results obtained in this paper can be explained by the interplay of two counteracting forces: (1) having fewer drivers increases service rate or, equivalently, reduces service time, and (2) having fewer drivers reduces the number of servers. In particular, we note the following (the statements is made precise in subsequent sections of the paper):

- Pickup time decreases with the number of idle vehicles. In a ride-hailing system, vehicles can be in one of three modes: (1) idle, (2) on the way to pick up a customer, and (3) en route transporting a customer (trip time). This means that the effective service time is the sum of pickup time and trip time. Whereas trip time is determined exogenously by the origin and

destination of the trips requested by the customers, pickup times depend on the number of idle vehicles. In particular, having fewer idle vehicles increases the likelihood that a customer request is matched with a distant vehicle, resulting in a longer pickup time. This means that, when the system is short on vehicles, pickup times are likely to be the longest.

- Having fewer drivers than vehicles reduces pickup times. Having fewer drivers than vehicles ensures that there are always idle vehicles with the fewer the drivers, the likelier it is for there to be more idle vehicles. Consequently, the fewer the drivers, the likelier it is that the pickup times are shorter (making overall shorter service times likelier). As we note next, having shorter service times comes, however, at the cost of having fewer servers.

- Having fewer drivers than vehicles reduces the number of servers. Although reducing the number of drivers shortens pickup times, it also reduces the number of servers available to handle customer requests. Hence, unless matched with a sufficient reduction in pickup times, reducing the number of drivers risks reducing overall service capacity.

Depending on the relative strength of these two forces, different outcomes are possible. Our analysis shows that this crucially depends on the number of vehicles relative to workload. In particular, the effect of shorter pickup times is strong when the likelihood of long pickup times is high. This is true when the system is short on vehicles. The effect of shorter pickup times is weak when the number of vehicles is large. In this case, the number of idle vehicles tends to be large, and hence, some reduction in the number of drivers does not have a significant effect.

These effects manifest themselves differently depending on whether customers are impatient or not. In the supply-limited regime, having fewer drivers can improve system throughput (amount of demand fulfilled) when customers are impatient. In the intermediate

regime, having fewer drivers can stabilize the system when customers are patient. That is, in the supply-limited (and intermediate) regimes, having fewer drivers paradoxically increases capacity. In the supply-rich regime, there is an opportunity to significantly reduce the number of drivers without significantly reducing the performance (as measured by throughput in the case of impatient customers and delay in the case of patient customers). In the paper, we make all of these statements precise and provide well-specified conditions under which the different noted outcomes arise.

Although we use ride-hailing as the motivating application for this paper, the models and analysis have broader applicability to other tele-operated services, such as food delivery (e.g., vehicles traveling to pick up a food order from a restaurant and then delivering to a home address) or warehouse operation (e.g., forklifts traveling to pick up products from one area of the warehouse to deliver them to another). The models and analysis can also be adapted to study settings in which the remote driver is needed for only one segment of the trip (e.g., the remote driver is needed to deliver the car to a customer, but the customer drives to the desired destination) and in which the vehicle may not be a car but a small robot that navigates over sidewalks instead.

The rest of the paper is organized as follows. In Section 2, we discuss related literature. In Section 3 (Section 4), we provide analysis for systems with impatient (patient) customers. In Section 5, we offer concluding comments. In the online appendices, we provide discussions of various extensions and proofs for all the formal results.

2. Related Literature

This paper contributes to the growing body of literature on on-demand mobility, including ride hailing. General reviews of this literature can be found in Benjaafar and Hu (2020), Hu (2021), and Freund et al. (2019). Papers within this literature that are of particular relevance include those that consider dimensioning problems and those that study spatial mismatches between supply and demand.

2.1. System Dimensioning

Besbes et al. (2022) consider the capacity-sizing problem for a ride-hailing service modeled as a multiserver queue. They show that the square root safety staffing rule (having buffer service capacity proportional to the square root of the nominal workload) is incapable of balancing the cost of supply capacity and service quality. Instead, buffer capacity must be in the order of a power of $\frac{2}{3}$ of the nominal workload. George and Xia (2011) model a one-way vehicle-sharing system as a closed queueing network and propose algorithms for determining the optimal number of vehicles. For the same problem,

Benjaafar et al. (2022) develop an approximation for the number of vehicles needed to guarantee a specified service level.

2.2. Spatial Mismatches

There is a growing body of literature that studies the inefficiencies of ride-hailing systems caused by spatial mismatches between demand and supply. The “wild goose chase” phenomenon, as described by Castillo et al. (2024), is the scenario in which, under limited supply, drivers are dispatched to pick up customers who are too distant, exacerbating the supply shortage. Feng et al. (2021) consider the case of a ride-hailing platform that operates on a circular road. They devise an approximation approach that models the nonmonotonic relationship between expected pickup time and customer arrival rate.

Additionally, there is a substantial body of research devoted to resolving spatial mismatches. A widely considered strategy is to impose a matching radius, meaning that customers and vehicles are only matched if they are sufficiently close; another strategy is to match a vehicle with the closest customer regardless of the order in which customers arrived (Xu et al. 2020, Feng et al. 2021, Wang et al. 2022, Castillo et al. 2024). Feng et al. (2021) propose a heuristic method for determining the near-optimal matching radius that minimizes customer average waiting time. Wang et al. (2022) develop a fluid model for a system in which a customer may cancel the service if the platform does not allocate a driver promptly. They obtain the fluid-based optimal matching radius in order to maximize system throughput. Xu et al. (2020) model the system as a double-ended queue and show that the smaller the matching radius, the weaker the wild goose chase effect. Castillo et al. (2024) demonstrate that, despite the fact that implementing the matching radius can resolve the wild goose chase, it is dominated by a surge-pricing strategy. In this paper, we show that the wild goose chase phenomenon could be mitigated using remote drivers who are fewer in numbers than vehicles.

Another stream of literature investigates spatial matching strategies. Banerjee et al. (2022b) adopt a closed queueing network model to study the dynamic matching between vehicles and customers. They propose a family of state-dependent policies that can result in an exponential decay of demand-loss probability as system parameters scale to infinity. Kanoria (2021) investigates matching units of supply and demand in a d -dimensional hypercube under various models, demonstrating how the minimum achievable cost (expected average distance between matched pairs) scales with model parameters and providing a matching algorithm that can achieve a near-optimal cost in the dynamic model. Özkan and Ward (2020) consider a ride-hailing system with time-varying driver and customer arrival rates, taking into consideration both the drivers' and

customers' patience. They propose a matching policy based on a continuous linear program and show that it is asymptotically optimal in terms of maximizing the number of (weighted) matches. There is also literature that studies how (temporally or spatially) pooling strategies, such as serving more than one customer at a time or allowing pickups only at designated locations) can be used to improve matching (see, for example, Santi et al. 2014, Cao and Qi 2022, Hu 2022, Chen and Hu 2023).

A stream of literature considers how curtailing demand, either indirectly via pricing and directly via admission control, can be used to address the spatial mismatch between supply and demand; see, for example, Wasserhole and Jost (2016), Banerjee et al. (2022a), Benjaafar and Shen (2023), and Afèche et al. (2023). The literature on admission control in queueing systems is extensive; see, for example, Stidham (1985, 2002), Altman et al. (2001), and Örmeci et al. (2001).

Our work contributes to the emerging body of literature on the impact of AVs on transportation. Benjaafar et al. (2023) investigate how a ride-hailing platform may use autonomous vehicles (through repositioning and vehicle assignment priorities) to achieve socially desirable outcomes. Siddiq and Taylor (2022), Lian and Van Ryzin (2024), Noh et al. (2021), and Castro and Frazelle (2022) examine the introduction of autonomous vehicles from a market design point of view, including the ownership structure and the competition between platforms. Mirzaeian et al. (2021) study the impact of AVs on highway traffic. They do so using a state-dependent queueing system. They do not consider the dynamics of passenger pickup and the associated spatial features. Our work also contributes to the stream of literature that considers service systems with a state-dependent service rate; see Delasay et al. (2019) for a recent review. In systems in which servers correspond to human operators, as in medical clinics or retail stores, there are empirical studies that provide evidence of servers either speeding up (see Kc and Terwiesch 2009, Wang and Zhou 2018 for example) or slowing down (see Batt and Terwiesch 2012, Chan et al. 2017) when these servers observe longer queues. Analytical models that incorporate state-dependent queues include Mandelbaum (1995), Mandelbaum and Pats (1998), Zhong et al. (2023), Delasay et al. (2016), and Dong et al. (2015).

Thematically, papers that are closest to ours are Hampshire et al. (2020) and Daw et al. (2024). Hampshire et al. (2020) consider a setting with autonomous vehicles in which remote drivers intervene only to resolve edge cases (i.e., situations that the automated system cannot handle on its own). They analyze the system using a standard multiserver queueing model (i.e., one in which the service times are not state dependent) and estimate the necessary number of remote drivers that would satisfy a target service using either the Erlang-B or Erlang-C formula. Daw et al. (2024) consider

a related system in which the requests for assistance from the remote drivers arrive in batches. Our work is different from Hampshire et al. (2020) and Daw et al. (2024) in that (1) we consider a setting in which the human support lasts for the full duration of the service, whereas Hampshire et al. (2020) and Daw et al. (2024) are primarily concerned with remote drivers handling edge cases, and (2) we account for the spatial feature of the service and how this feature interacts with the number of servers. As a result, the modeling and analysis are different, and so are the results.

Finally, there is growing literature in various fields of engineering that addresses the technical requirements for tele-operated driving. Reviews of this literature can be found in Zhang (2020) and the references therein.

3. Problem Formulation

We consider a ride-hailing platform that relies on remote drivers who operate vehicles at a distance. Requests for trips arise continuously over time with each request associated with a pickup and destination location across a specified service region. Let m and n , where $m \geq n$, denote the number of vehicles and remote drivers, respectively (when $n = m$, the system with remote drivers, all else being the same, has dynamics similar to those of a system with in-vehicle drivers).²

We consider the case in which customers are impatient (the case of patient customers is treated in Section 4). That is, if there are no idle drivers when a customer makes a trip request, the request cannot be fulfilled, and the customer is considered lost. If there are idle drivers (which also implies the presence of idle vehicles because $m \geq n$) at the time a trip request is made, an idle driver is assigned to the vehicle that is nearest to the origin of the customer request. The driver takes over the control of the vehicle and drives it (remotely) to the location of the customer. Once the customer is picked up, the remote driver drives the vehicle to the customer's requested destination. Upon trip completion, the driver parks the vehicle where the service is terminated, and the driver and the vehicle are uncoupled and become available to be independently assigned to future customer requests.³

The total service time for a customer who is matched with a vehicle and a driver consists of two components: a pickup time and a trip time. That is, total service time = pickup time + trip time. The pickup time corresponds to the time it takes the idle vehicle nearest to the customer to travel from its current location to the location of the customer. The trip time corresponds to the time it takes the vehicle to travel from the customer's current location (the trip origin) to the customer's requested destination (the trip destination). Because the pickup time depends on the location of the nearest vehicle, it varies with the number of idle vehicles. For example, when the number

of idle vehicles is large, pickup time is more likely to be short, whereas the reverse is true (pickup time is more likely to be long when the number of idle vehicles is small). In other words, pickup time (and, consequently, total service time) is dependent on the state of the system.

The system as described can be viewed as a special case of a multiserver queueing system with state-dependent service times. An exact treatment of this system is difficult in general as service times are dependent on the demand process, its spatial distribution, and the topology of the road network. To allow for tractability, we ground our analysis in a Markovian approximation of both the demand and service processes. In particular, we approximate the demand process by a Poisson process with rate λ , where the origin–destination pairs associated with each requested trip are assumed to be uniformly distributed over the service region.⁴ The service region is approximated by a continuous area, denoted by $\mathcal{C}$, that is a bounded and convex subset of $\mathbb{R}^2$. We also approximate the distribution of service times by a state-dependent exponential distribution⁵ with rate $\mu(m, q)$, where $q \in \{1, \dots, n\}$ is the number of customers currently being served (customers waiting to be picked up + customers en route to their destination) and m is the number of vehicles. Note that, for a given m , $\mu(m, q)$ is a function of q defined for $q \in [1, m]$. We assume that $\mu(m, q)$ satisfies a few mild conditions stated subsequently that account for the spatial feature of a ride-hailing system.

Assumption 1. We assume the following holds:

1. $\mu(m, q)$ is strictly increasing in m and decreasing in q .
2. $\mu(m, q)$ is strictly concave in q .
3. $\lim_{m \rightarrow \infty} \mu(m, q) = 1/s$, where s is the expected trip between two uniformly⁶ drawn locations in $\mathcal{C}$.
4. $\mu(m, m)$ is invariant in m .⁷

Assumption 1(1) specifies that the service rate is increasing in the number of vehicles and decreasing in the number of customers currently being served or, equivalently, is increasing in the number of idle vehicles. Assumption 1(2) states that the decrease in the service rate because of having one fewer idle vehicle is more pronounced when the number of idle vehicles is low. Assumption 1(3) states that the pickup time approaches zero as the number of vehicles approaches infinity. Assumption 1(4) states that the service rate does not vary with the number of vehicles when there is no idle vehicle present.

To account for the trade-off between reduced service capacity and shorter service times, we make the following assumption about the rate at which trips are completed when there are q customers in service, $q\mu(m, q)$.

Assumption 2. There exists $\bar{m} > 0$ such that, if $m \geq \bar{m}$,⁸ $q\mu(m, q)$ first increases then decreases in q .

Assumption 2 states that the rate at which trips are completed when there are q customers in service,

$q\mu(m, q)$, is not monotonically increasing in the number of customers being served or, equivalently, with the number of busy drivers. Note that $q\mu(m, q)$ is the product of two terms: one increasing, q , and the other decreasing, $\mu(m, q)$. The nonmonotonicity implies that the effect of q is stronger when q is small, whereas the effect of $\mu(m, q)$ is stronger when q is large. In Online Appendix H, we show that Assumptions 1 and 2 hold for a wide range of service region geometries and are consistent with various approximations of the functional form of $\mu(m, q)$ that have been employed in the literature (see, for example, Feng et al. 2021, Kanoria 2021, Besbes et al. 2022). Note that, in a standard multiserver queue in which the service rate is not state-dependent (i.e., one in which $q\mu(m, q) = q\mu$), $q\mu(m, q)$ is monotonically increasing in q .

Using Kendall's notation, the queueing system we consider is an $M/M(q)/l/l^q$ queueing system (or an Erlang loss system with a state-dependent service rate) in which l is the number of servers (for our system with m vehicles and n driver, $l = n$).

Let $\pi_{m,n}(q)$ denote the stationary probability of having q customers in the system given that the system has m vehicles and n remote drivers. These probabilities can be obtained by solving the underlying Markov chain, a birth and death process with birth rate λ and death rate $q\mu(m, q)$ when the system is in state q . Letting

$$\rho(q) = \frac{\lambda}{q\mu(m, q)}, \quad (1)$$

we have

$$\pi_{m,n}(0) = \left[1 + \sum_{i=1}^n \prod_{k=1}^i \rho(k) \right]^{-1}, \quad \text{and} \quad (2)$$

$$\pi_{m,n}(q) = \pi_{m,n}(0) \prod_{k=1}^q \rho(k), \quad \text{for } 1 \leq q \leq n. \quad (3)$$

Let $SL(m, n)$ denote the service level for a system with m vehicles and n remote drivers, in which service level is defined as the long-run fraction of demand fulfilled. Noting that customers are turned away when all drivers are busy, we have $SL(m, n) = 1 - \pi_{m,n}(n)$.

We conclude this section by highlighting two important features of systems with remote drivers (i.e., systems in which $n < m$). First, note that for a system with $n < m$, the service rate $\mu(m, q)$ when there are q customers in the system is the same as for one with $n = m$ and q customers in the system. That is, even though there are fewer drivers than vehicles, a system with remote drivers can leverage the larger number of vehicles to maintain the same service rate. This suggests that reducing the number of drivers when the drivers are remote may have a lesser impact on performance than in a traditional system in which reducing the number of drivers

is accompanied by a reduction in the number of vehicles.

Second, note that the dynamics of the underlying birth and death process are the same as those for a system with $n = m$ except that the state space is truncated at $q = n$ instead of $q = m$. This means that customers are rejected when the system is in state $q = n$ instead of $q = m$. However, by preventing the system from being in states higher than n , expected service times are upper bounded by $1/\mu(m, n)$ (instead of $1/\mu(m, m)$ when $n = m$). The increased likelihood of shorter service times (because of always having at least $m - n$ idle vehicles) can be particularly valuable when the system is highly loaded and the likelihood of states higher than $q = n$ is high. In other words, by limiting the number of drivers, the likelihood of shorter service time is increased, albeit at the expense of decreasing the number of servers.

In view of these features, it is reasonable to conjecture that, (a) when the system is highly loaded, having fewer drivers can actually lead to a better service level (because of the increased likelihood of shorter service times), and (b) when the system is lightly loaded, it may be possible to reduce the number of drivers without significantly reducing service level. The rest of this section provides confirmation for both conjectures.

3.1. Preliminaries: A System with an Equal Number of Vehicles and Drivers ($n = m$)

Before we provide additional analysis for systems with remote drivers in which $n \leq m$, let us consider, as a benchmark, a system with $n = m$ (all else being the same, this can be viewed as a system with in-vehicle drivers). As we show in the following proposition, there are three distinct regimes of operation, depending on the number of vehicles, characterized by differences in the features of the distribution of the number of customers in the system and the associated system performance. In the next section, we describe how the introduction of remote drivers (i.e., allowing for the possibility of $n < m$) affects these features and the associated system performance. Recall that $\bar{m}$ is introduced in Assumption 2. Throughout the remainder of this paper and for ease of presentation, we restrict ourselves to the case in which $m \geq \bar{m}$ and $\lambda > \max\left\{\frac{1}{s}, \max_{q \in \{1, 2, \dots, \bar{m}\}} q\mu(\bar{m}, q)\right\}$, which implies that the system being studied is not too small in size¹⁰ (in the setting we consider, both $\bar{m}$ and the lower bound for λ are very small).

Proposition 1. *There exist $\gamma_1(\lambda)$ and $\gamma_2(\lambda)$ with $0 < \gamma_1(\lambda) < \gamma_2(\lambda)$ such that the probabilities $\pi_{m,m}(q)$ satisfy the following properties:*

- When $m < \gamma_1(\lambda)$, $\pi_{m,m}(q)$ increases in q for $q \leq m$, implying that $\pi_{m,m}(q)$ is unimodal with the mode at m .
- When $\gamma_1(\lambda) < m < \gamma_2(\lambda)$, there exist q_1 and q_2 with $0 < q_1 < q_2$ such that $\pi_{m,m}(q)$ increases in q for $q < q_1$, decreases in q for $q_1 < q < q_2$, and then increases in q for

$q > q_2$, implying that $\pi_{m,m}(q)$ is bimodal with one mode at $\lfloor q_1 \rfloor < m$ and the other at m .

- When $m > \gamma_2(\lambda)$, there exists $q_3 > 0$ ¹¹ such that $\pi_{m,m}(q)$ increases in q for $q < q_3$ and decreases in q for $q > q_3$, implying that $\pi_{m,m}(q)$ is unimodal with the mode at $\lfloor q_3 \rfloor < m$.

Proposition 1 shows that, depending on the number of vehicles m relative to the customer arrival rate λ , the system can be in one of three regimes: (1) $m < \gamma_1(\lambda)$ (we refer to this regime as the supply-limited regime), (2) $\gamma_1(\lambda) < m < \gamma_2(\lambda)$ (we refer to this regime as the intermediate regime), and (3) $m > \gamma_2(\lambda)$ (we refer to this regime as the supply-rich regime).

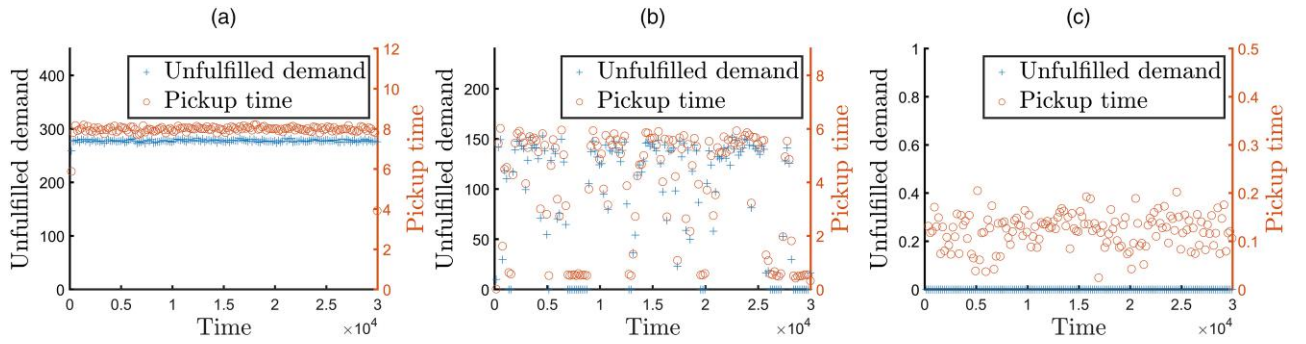
In the supply-limited regime, the probability $\pi_{m,m}(q)$ is increasing in q . This implies that the more likely states are those in which the system is critically loaded with the number of customers in the system being close to the number of vehicles. An arriving customer is most likely either to be rejected or experience a long pickup time. In the intermediate regime, the distribution of the number of customers in the system is bimodal with one mode at $\lfloor q_1 \rfloor < m$ and the other at m . The number of customers oscillates between these two modes, suggesting different experiences for customers at different times. Customers who arrive during periods when the system state is around $\lfloor q_1 \rfloor$ are served and experience relatively short pickup times. Customers who arrive during periods when the number of customers in the system is close to m are either rejected or experience long pickup times. In the supply-rich regime, the distribution of the number of busy vehicles is unimodal with the mode at $\lfloor q_3 \rfloor < m$. In this regime, customers are more likely to be served and experience relatively short pickup times.

Remark 1. Depending on the functional form of $\mu(m, q)$, it may be possible to obtain explicit expressions for $\gamma_1(\lambda)$ and $\gamma_2(\lambda)$ with $0 < \gamma_1(\lambda) < \gamma_2(\lambda)$. For example, in Online Appendix D (see (13)), we show that this is the case when

$$\mu(m, q) = \left[\frac{s}{\sqrt{m - q + 1}} + s \right]^{-1}. \quad (4)$$

This functional form of service rate is adopted by Besbes et al. (2022), who show that it is asymptotically exact for a setting similar to ours; see also Section 3.3 for additional related discussion and details.

To illustrate the temporal dynamics in each of the regimes (and to validate the intuition), we provide simulation results in Figure 2 for an example system in which the customer arrival rate is $\lambda = 500$ per minute and the state-dependent service rate is given by (4) with $s = 10$. We simulate the corresponding $M/M(q)/I/I$ queueing system,¹² starting in state 0 (i.e., starting with an empty system with no customers in service). The simulation generates sample paths of customer

Figure 2. (Color online) The Impact of Vehicle Supply Capacity on Pickup Times and Unfulfilled Demand

Notes. (a) $m = 4,000$. (b) $m = 5,600$. (c) $m = 12,000$.

demand not fulfilled (averaged every 200 minutes) and pickup time (averaged every 200 minutes) over a time period of 30,000 minutes. Panels (a)–(c) illustrate, respectively, the temporal dynamics for a supply-limited regime ($m = 4,000$), an intermediate regime ($m = 5,600$), and a supply-rich regime ($m = 12,000$). As we can see, in the supply-limited regime (panel (a)), both pickup time and unfulfilled demand are, over time, consistently high. In the intermediate regime (panel (b)), pickup time and unfulfilled demand oscillate between high and low values.¹³ In the supply-rich regime (panel (c)), both pickup times and unfulfilled demand are negligible.

Besbes et al. (2022) make related observations for a similar system (see proposition 1 in Besbes et al. (2022)). Our results are different in that (1) we focus on finite systems, whereas Besbes et al. (2022) analyze systems in heavy traffic; (2) we consider systems with impatient customers, whereas they consider systems with patient customers; and (3) we characterize different regimes by vehicle capacity, whereas they use scaling parameters that capture the rate under which the nominal workload relative to the server capacity approaches one.

3.2. The System with Fewer (Remote) Drivers Than Vehicles ($n \leq m$)

In this section, we provide analysis for the case with remote drivers (i.e., $n \leq m$) and examine how the introduction of remote drivers, when drivers may be fewer than vehicles, impacts system dynamics and performance. We begin by describing how the results in Proposition 1 (the features of the distribution of the number of customers) are affected. Recall that $\gamma_1(\lambda)$, $\gamma_2(\lambda)$, q_1 , q_2 , and q_3 are introduced in Proposition 1.

Proposition 2. *The probabilities $\pi_{m,n}(q)$ satisfy the following properties:*

i. When $m < \gamma_1(\lambda)$, $\pi_{m,n}(q)$ increases in q for $q \leq n$, implying that $\pi_{m,n}(q)$ for $q \in \{0, \dots, n\}$ is unimodal with the mode at n .

ii. When $\gamma_1(\lambda) < m < \gamma_2(\lambda)$, depending on n , we have the following:

ii.i. If $n < q_1$, $\pi_{m,n}(q)$ increases in q for $q \leq n$, implying that $\pi_{m,n}(q)$ is unimodal with the mode at n .

ii.ii. If $q_1 < n < q_2$, $\pi_{m,n}(q)$ increases in q for $q < q_1$ and decreases in q for $q_1 < q \leq n$, implying that $\pi_{m,n}(q)$ is unimodal with the mode at $\lfloor q_1 \rfloor$.

ii.iii. If $n > q_2$, $\pi_{m,n}(q)$ increases in q for $q < q_1$, decreases in q for $q_1 < q < q_2$, and then increases in q for $q_2 < q \leq n$, implying that $\pi_{m,n}(q)$ is bimodal with one mode at $\lfloor q_1 \rfloor$ and the other at n .

iii. When $m > \gamma_2(\lambda)$, depending on n , we have the following:

iii.i. If $n < q_3$, $\pi_{m,n}(q)$ increases in q for $q \leq n$, implying that $\pi_{m,n}(q)$ is unimodal with the mode at n .

iii.ii. If $n > q_3$, $\pi_{m,n}(q)$ increases in q for $q < q_3$ and decreases in q for $q_3 < q \leq n$, implying that $\pi_{m,n}(q)$ is unimodal with the mode at $\lfloor q_3 \rfloor$.

Proposition 2 shows that, depending on the number of vehicles, the system can again be in one of three similarly defined regimes (i.e., the numerical values of the thresholds on vehicle capacity are the same) as those defined in Proposition 1 for a system with an equal number of vehicles and drivers. We again refer to these three regimes as supply-limited, intermediate, and supply-rich regimes. Although the regimes are similarly defined, there are important differences in the features of the distribution of the number of customers (or, equivalently, busy vehicles) in each regime.

In the supply-limited regime, the distribution of the number of customers in the system is again unimodal. However, the mode is now at n . The more likely states are those in which the number of customers in the system is close to the number of drivers. However, pickup times in those states are now shorter because the number of vehicles available is always greater than or equal to $m - n$ (which is strictly positive when $n < m$). Because pickup times are shorter, this suggests that enough driver capacity may be freed up (as long as the number

of drivers is not too small) to allow for more demand to be fulfilled.

In the intermediate regime, we observe three subregimes with respect to n . If the ratio of driver capacity to vehicle capacity is low, the distribution of the number of customers in the system is unimodal with the mode at n . If the ratio of driver capacity to vehicle capacity is moderate, the distribution of the number of customers in the system is unimodal with the mode at $\lfloor q_1 \rfloor < n$. If the ratio of driver capacity to vehicle capacity is high, the distribution of the number of customers in the system is bimodal with one mode at $\lfloor q_1 \rfloor$ and the other at n . Therefore, compared with Proposition 1, having remote drivers that are fewer than vehicles can change the number of modes.

In the supply-rich regime, we observe two subregimes with respect to n . If the ratio of driver capacity to vehicle capacity is low, the distribution of the number of customers in the system is unimodal with the mode at n . If the ratio of driver capacity to vehicle capacity is high, the distribution of the number of customers in the system is unimodal with the mode at $\lfloor q_3 \rfloor$. In the supply-rich regime, having fewer drivers can shift the mode of the distribution from $\lfloor q_3 \rfloor$ to n . However, as long as $n > \lfloor q_3 \rfloor$, the more likely states are those around $\lfloor q_3 \rfloor$. Because the service rate only depends on the number of idle vehicles in the system (as long as there are some idle drivers), this suggests that it may be possible to reduce the number of drivers without significantly impacting system performance.

To illustrate the temporal dynamics in each of the regimes (and to validate the intuition), we provide simulation results in Figure 3 using the same parameters and procedure as those used for the results shown in Figure 2 for a system with an equal number of vehicles and drivers. Panels (a)–(c) illustrate, respectively, the temporal dynamics for a supply-limited regime ($m = 4,000$ and $n = 3,500$), an intermediate regime ($m = 5,600$ and $n = 5,250$), and a supply-rich regime ($m = 12,000$ and $n = 8,000$). As we can see, in the supply-limited

regime, the system with remote drivers loses less demand, and the pickup time is shorter (panel (a) in Figure 3) than a system with an equal number of vehicles and drivers (panel (a) in Figure 2). In the intermediate regime, the system with remote drivers eliminates the oscillation patterns (panel (b) in Figure 2) and has consistently lower pickup time and unfulfilled demand (panel (b) in Figure 3). In the supply-rich regime, the system with remote drivers maintains a similar pickup time and level of unfulfilled demand (panel (c) in Figure 3) as the system with an equal number of vehicles and drivers (panel (c) in Figure 2).

We next present a set of results (Theorems 1–3) regarding service level in a system with remote drivers. Recall that we use $SL(m, n)$ to denote the service level in a system with m vehicles and n remote drivers.

Theorem 1. For $m < \gamma_1(\lambda)$, if

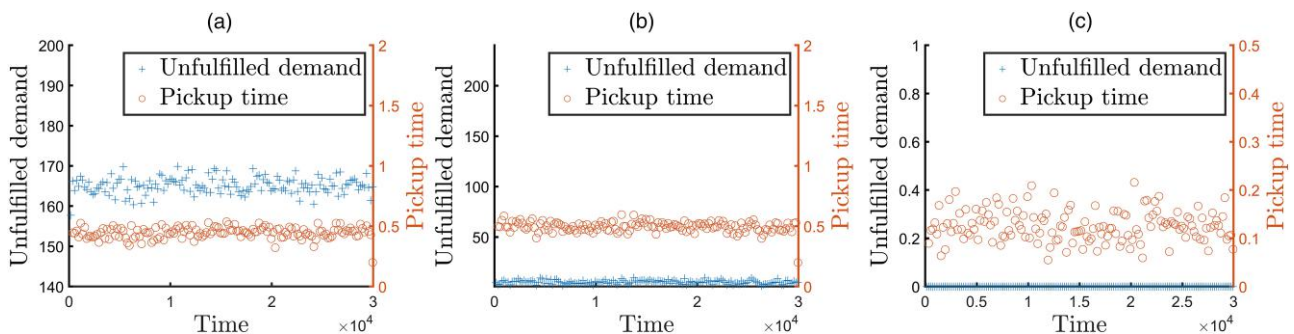
$$\lambda SL(m, m) > m\mu(m, m), \quad (5)$$

then there exists a threshold $\tilde{n}_1 \leq m - 1$ such that $SL(m, n)$ is increasing in n when $n \leq \tilde{n}_1$ and is decreasing in n when $n > \tilde{n}_1$. Otherwise, $SL(m, n)$ is nondecreasing in n .

Theorem 1 shows that, under the well-specified Condition (5) (more about this condition later), service level may not be monotonic in the number of drivers. More importantly, Theorem 1 shows that conditions exist under which reducing the number of drivers improves service level (i.e., having fewer drivers can improve service level). The fact that the service level may not be monotonic in the number of drivers n implies that there is an optimal number of drivers under which the service level is maximized (this optimal number of drivers can be obtained via a simple line search).

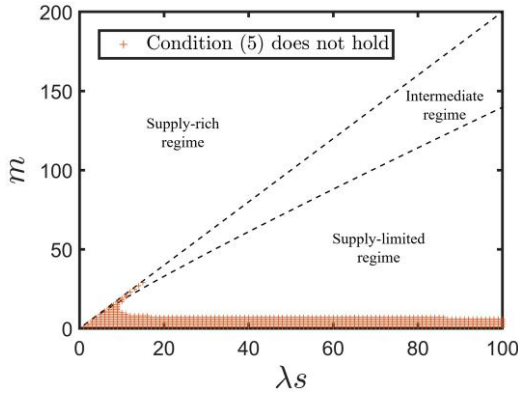
Condition (5) can be elucidated as follows. The right-hand side of the inequality represents the rate at which trips are completed when all drivers are occupied in a system with an equal number of drivers and vehicles, whereas the left-hand side represents, for the same system, the rate at which trip requests are fulfilled.

Figure 3. (Color online) The Impact of Vehicle Supply Capacity and Remote Driver Capacity on Pickup Times and Unfulfilled Demand



Notes. (a) $m = 4,000$, $n = 3,500$. (b) $m = 5,600$, $n = 5,250$. (c) $m = 12,000$, $n = 8,000$.

Figure 4. (Color online) An Illustration of the Parameter Range for Condition (5) to Hold in the Supply-Limited and Intermediate Regimes



Condition (5) captures the trade-off between reduced service capacity and shorter service times. Note that in a standard multiserver queue in which the service rate is not state-dependent (i.e., one in which $q\mu(m, q) = q\mu$), the inequality does not hold. This implies that Assumption 2 (which states that $q\mu(m, q)$ first increases then decreases in q) is pivotal to the result that reducing the number of drivers relative to the number of vehicles can actually improve service level.

In Figure 4, we consider an example system in which the state-dependent service rate is given by (4) (i.e., $\mu(m, q) = [(s/\sqrt{m-q+1}) + s]^{-1}$). In this case, Condition (5) reduces to one that depends only on the number of vehicles m and the system workload λs . Figure 4 illustrates the range of parameter values under which the condition holds. The results suggest that (5) holds in the supply-limited regime as long as m is not too small.

To illustrate the efficiencies gained by operating a system with remote drivers, we provide in Figure 5

numerical results for an example system in which $\mu(m, q) = [(s/\sqrt{m-q+1}) + s]^{-1}$, $\lambda = 10$, and $s = 10$. We vary the vehicle capacity (m) and, in each case, choose the number of drivers that maximizes the service level. Panel (a) shows both the percentage and absolute (relative to the benchmark system in which $n = m$) improvement in service level as the number of vehicles is varied. Panel (b) shows both the percentage and absolute reduction in the number of drivers as the number of vehicles is varied. In Section 3.3, we provide sharper analytical results by considering the asymptotic regime of very large demand.

Theorem 2 provides an analogous result to Theorem 1 for the intermediate regime.

Theorem 2. For $\gamma_1(\lambda) < m < \gamma_2(\lambda)$, if (5) holds, then there exists a threshold $\tilde{n}_2 \leq m - 1$ such that $SL(m, n)$ is increasing in n when $n \leq \tilde{n}_2$ and is decreasing in n when $n > \tilde{n}_2$. Otherwise, $SL(m, n)$ is nondecreasing in n .

Theorem 2 shows that, for the intermediate regime, Condition (5) is again sufficient and necessary for service level to be nonmonotonic in the number of drivers and for reducing the number of drivers to improve service level. Figure 4 shows that Condition (5) holds (when $\mu(m, q) = [(s/\sqrt{m-q+1}) + s]^{-1}$) in the intermediate regime as long as either m or λs is not too small. Figure 6 provides numerical results (for the same example considered for Figure 5) illustrating gains that can be obtained by using remote drivers (relative to a system with $n = m$). The results show that these gains are less significant compared with those observed in the supply-limited regime. In Section 3.3, we provide analytical support for these observations.

For the supply-rich regime, we have the following result.

Theorem 3. For $m > \gamma_2(\lambda)$, $SL(m, n)$ is increasing in n . Moreover, for any $n > q_3$, where q_3 is introduced in Proposition 1, $SL(m, n)$ is lower bounded by $1 - (1/n - \lfloor q_3 \rfloor)$.

Figure 5. (Color online) The Impact of Using Remote Drivers on System Performance in the Supply-Limited Regime

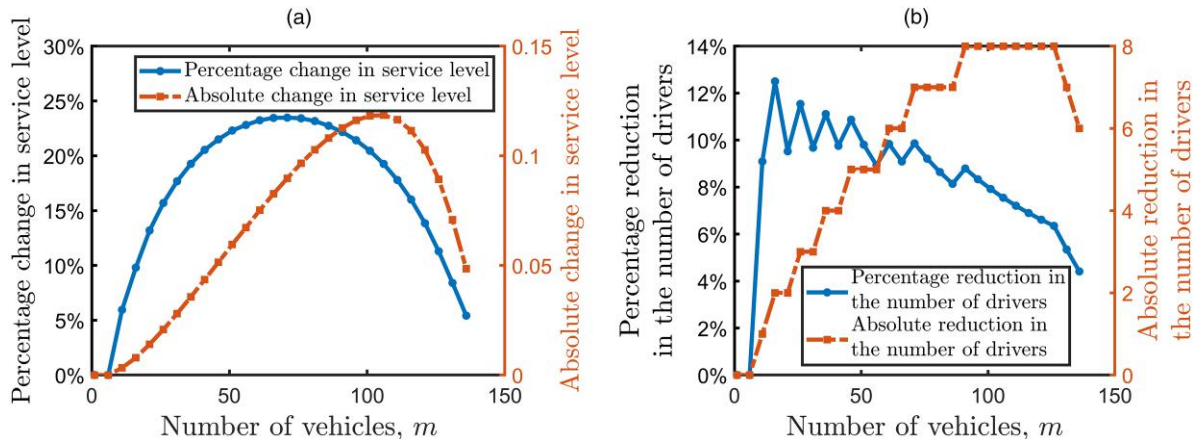
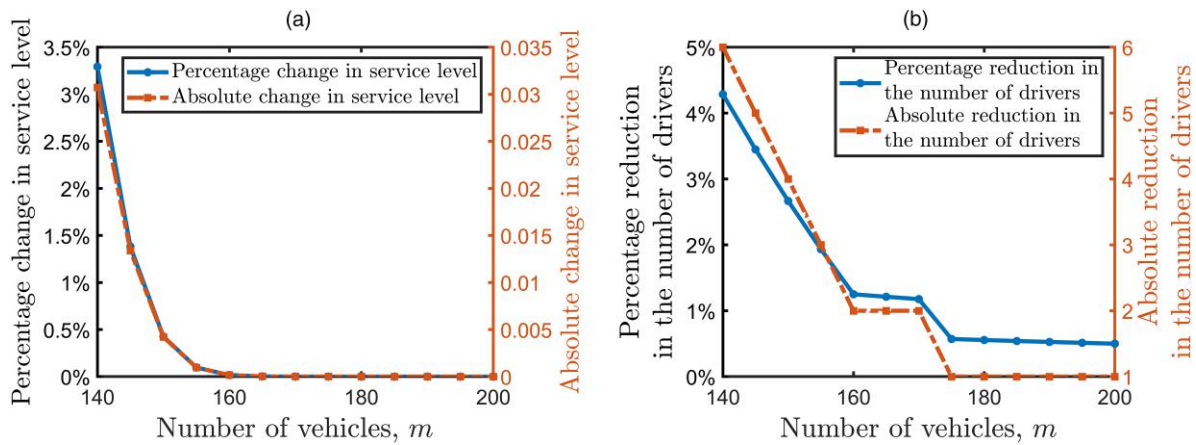


Figure 6. (Color online) The Impact of Using Remote Drivers in the Intermediate Regime



The first part of Theorem 3 shows that, in the supply-rich regime, reducing the number of drivers always reduces the service level. The second part of the theorem shows that, when the number of drivers is sufficiently large (i.e., $n > q_3$), the service level is lower bounded by a function that is increasing concave in n and has the form $1 - (1/n - \lfloor q_3 \rfloor)$. This suggests that the number of drivers may have a diminishing effect and there may be an opportunity to reduce the number of drivers without significantly reducing the service level. This observation is supported by the numerical results in Figure 7. In the next section, we provide analytical support for these observations.

3.3. Asymptotic Analysis

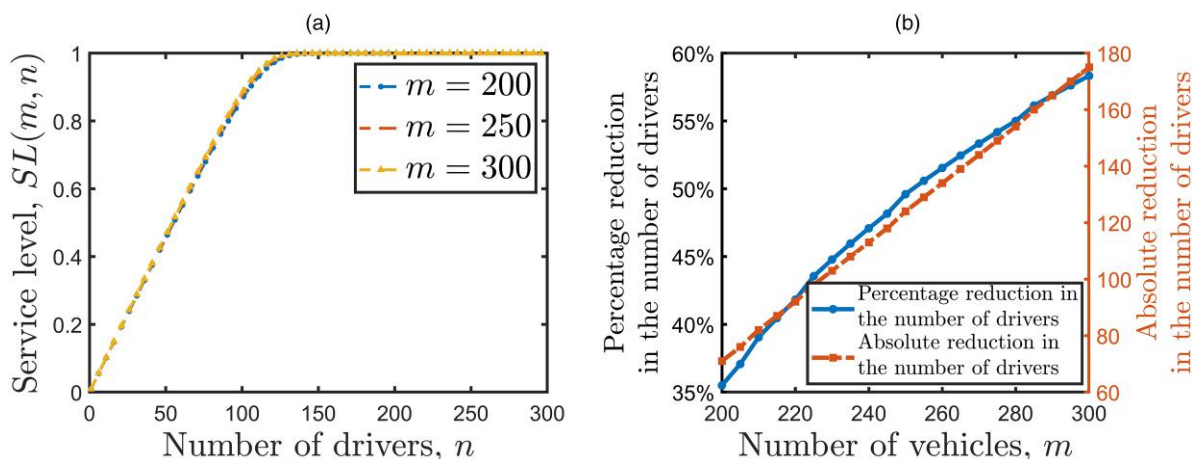
In this section, we consider the asymptotic regime in which the demand approaches infinity and the number of vehicles scales proportionally. Such a setting, corresponding to a large market, may be of particular

relevance to a service that operates in a large, dense, urban area. In what follows, we show that, in such a setting, it is possible to provide a sharper characterization of the impact of using remote drivers on service level.

To carry out this analysis, we rely on the asymptotically optimal approximation of service rate proposed by Besbes et al. (2022) referenced in Section 3.1: $1/\mu(m, q) = (s/\sqrt{m - q + 1}) + s$, where s corresponds the expected travel time between two uniformly drawn locations in the service region $\mathcal{C}$. The term $s/\sqrt{m - q + 1}$ captures the pickup time, and the term s captures trip time. Note that this approximation of service rate satisfies Assumptions 1 and 2. Moreover, as mentioned earlier, it is asymptotically exact (up to a geometry-dependent coefficient) when the origin–destination pairs associated with each requested trip is uniformly distributed over $\mathcal{C}$ (see lemma 1 in Besbes et al. (2022) for validation and detailed discussions).

We provide results for each of the supply regimes considered in the previous sections. Note that, because

Figure 7. (Color online) The Impact of Using Remote Drivers in the Supply-Rich Regime



Note. Panel (b) shows both the maximum percentage and maximum absolute reductions in the number of drivers that guarantee a reduction in service level of less than 0.01 (relative to a system with $n = m$).

we consider proportional scaling of demand and number of servers such that $m_\lambda = \lfloor \alpha \lambda s \rfloor$ for $\alpha > 0$, the conditions that correspond to different supply regimes have now much simpler forms. In particular, because $\lim_{\lambda \rightarrow \infty} \lambda s / \gamma_1(\lambda) = 1$ and $\lim_{\lambda \rightarrow \infty} \lambda s / \gamma_2(\lambda) = 1/2$, the system is in the supply-limited regime if $\alpha \in (0, 1)$, the intermediate regime if $\alpha \in (1, 2)$, and the supply-rich regime if $\alpha > 2$ (see Online Appendix D for detailed analysis).

We consider first the supply-limited regime.

Theorem 4. Let $m_\lambda = \lfloor \alpha \lambda s \rfloor$ for $\alpha \in (0, 1)$, then $\liminf_{\lambda \rightarrow \infty} [\max_{n \in \{1, \dots, m_\lambda\}} SL(m_\lambda, n) / SL(m_\lambda, m_\lambda)] \geq 2 - \alpha$.

Theorem 4 shows that, in the supply-limited regime, a system with remote drivers (when the number of drivers is chosen to maximize service level) improves service level (relative to a system with an equal number of drivers and vehicles) asymptotically by at least a factor of $2 - \alpha$ ($\alpha \in (0, 1)$). The bound on the improvement decreases in α and vanishes as α approaches one, which is consistent with the result we obtain for the intermediate regime. In Online Appendix D.4, we provide numerical results comparing the asymptotic bound and the service level ratio derived from the simulation of a finite system.

Theorem 5. Let $m_\lambda = \lfloor \alpha \lambda s \rfloor$ for $\alpha \in (1, 2)$, then $\lim_{\lambda \rightarrow \infty} [\max_{n \in \{1, \dots, m_\lambda\}} SL(m_\lambda, n) / SL(m_\lambda, m_\lambda)] = 1$.

Theorem 5 shows that, in the intermediate regime, the relative difference between a system with remote drivers (with the number of drivers chosen optimally to maximize service level) and a system with an equal number of drivers and vehicles vanishes asymptotically. The result in Theorem 5 can be explained by the fact that, as λ approaches infinity, the more likely states (in a system with equal numbers of drivers and vehicles) are those in which the system is noncritically loaded (states around $\lfloor q_1 \rfloor$). In fact, in this case $\pi_{m_\lambda, m_\lambda}(\lfloor q_1 \rfloor)$ dominates (with a higher order) $\pi_{m_\lambda, m_\lambda}(m_\lambda)$.

Finally, we consider the supply-rich regime in Theorem 6.

Theorem 6. Let $m_\lambda = \lfloor \alpha \lambda s \rfloor$ for $\alpha > 2$ and $n_\lambda^* = \min\{n : SL(m_\lambda, m_\lambda) - SL(m_\lambda, n) < \epsilon\}$ for $\epsilon > 0$. Then, $\limsup_{\lambda \rightarrow \infty} n_\lambda^* / m_\lambda \leq 1/\alpha$.¹⁴

Theorem 6 shows that, under the supply-rich regime, a system with remote drivers can reduce the number of drivers (relative to a system with an equal number of drivers and vehicles) asymptotically by at least a factor of $\frac{1}{\alpha}$ (ϵ can be arbitrarily small). Because $\alpha > 2$ in this case, this means that the number of drivers can be reduced by at least half. In Online Appendix D.4, we provide numerical results comparing the asymptotic bound and the actual driver-to-vehicle ratio derived from the simulation of a finite system.

3.4. Numerical Results Using Data from New York City

In this section, we provide numerical results in which the demand process, pickup times, and trip times are determined based on real-world data. The data we use is from the New York City Taxi & Limousine Commission (TLC).¹⁵ The data contains GPS coordinates for the pickup and drop-off locations of all yellow cab trips over multiple years and the associated pickup and drop-off times (note that the pickup time in the TLC data refers to the time when the customer is picked up by the taxi, which is different from the pickup time defined in this paper). Because information about pickup times, as defined in this paper, is not provided in the TLC data, we estimated these times using the city's road network and the prevailing travel speeds for the times considered. The data set we use covers the periods of June 2015 to August 2015. More information on the data and the numerical procedure can be found in Online Appendices G.1–G.2.

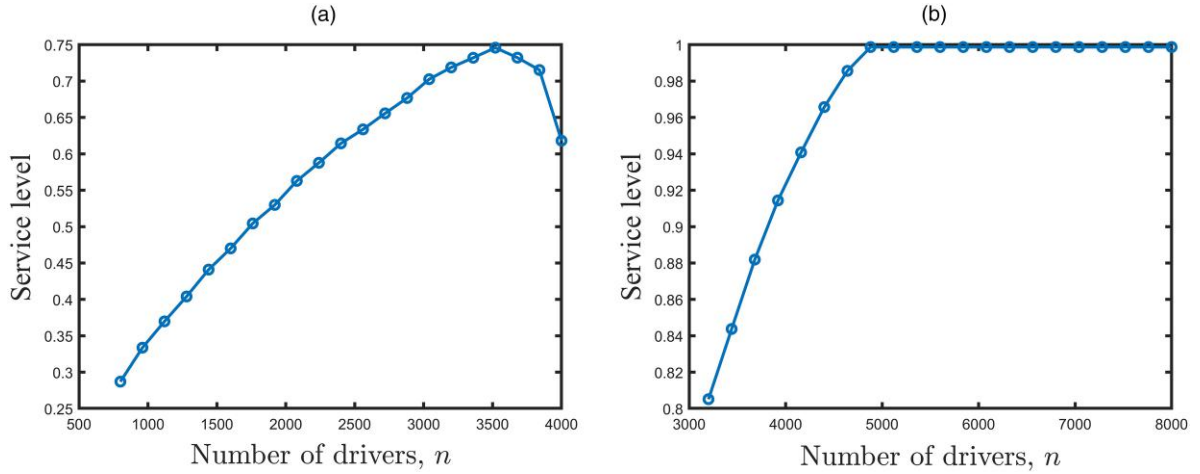
Figure 8 provides representative results (based on data from June 16, 2015). In particular, results for two scenarios for the number of vehicles are shown, $m = 4,000$ (panel (a)) and $8,000$ (panel (b)), corresponding, respectively, to the supply-limited and supply-rich regimes. The results are consistent with those in Section 3.2. In the supply-limited regime, a system in which the number of remote drivers is appropriately selected can improve the service level by 20.7%. In the supply-rich regime, a system with remote drivers can reduce the number of drivers by 42% (relative to the number of vehicles), maintaining roughly the same service level. Additional numerical results can be found in Online Figure G.3. A more extensive set of results is also available upon request.

4. Systems with Patient Customers

In this section, we consider the case in which customers are patient. That is, we consider the setting in which customers are willing to wait until both a vehicle and driver are available to serve them. We assume that customers are served on a first come, first served basis and are matched with the closest idle vehicle, a so-called first dispatch policy (see, for example, Castillo et al. 2024).

The total amount of time a customer spends in the system has now three components: (1) time waiting for a vehicle and driver to be dispatched, (2) time it takes a vehicle to pick up a customer once it is dispatched, and (3) time it takes a vehicle to complete the trip (from the pickup location to the requested destination) with the customer on board. We refer to the first component as waiting time and continue to refer to the second and third as pickup time and trip time, respectively. Hence, the customer's total time in the system is now the sum waiting time + pickup time + trip time.

Figure 8. (Color online) Results from Numerical Experiments Based on TLC Data



Notes. (a) Supply-limited regime. (b) Supply-rich regime.

As with the case of impatient customers, we approximate the dynamics of the system by those of a multiserver queueing system with state-dependent service time but now with an infinite waiting room, namely, an $M/M(q)/n$ queue. We assume that the state-dependent service rate function, $\mu(m, q)$, satisfies Assumptions 1 and 2.

Recall that we define $\rho(q) = \lambda/q\mu(m, q)$. A condition for the system to be stable (a queueing system is said to be stable if its long-run average over time of the number of customers in the system exists and is finite) is given by the following lemma. Note that stability implies a finite expected delay.

Lemma 1. *The system is stable if and only if $\rho(n) < 1$.*

Let $\pi_{m,n}(q)$ denote the stationary probability of having q customers in the system given that the system has m vehicles and n remote drivers. These probabilities can be obtained by solving the underlying Markov chain, a birth and death process with birth rate λ and death rate $\min\{q, n\}\mu(m, \min\{q, n\})$, when the system is in state q . Given that the stability condition in Lemma 1 is satisfied,

$$\pi_{m,n}(0) = \left[1 + \sum_{i=1}^n \prod_{k=1}^i \rho(k) + \frac{\rho(n)}{1 - \rho(n)} \prod_{k=1}^n \rho(k) \right]^{-1}, \quad (6)$$

$$\pi_{m,n}(q) = \pi_{m,n}(0) \prod_{k=1}^q \rho(k), \text{ for } q \in \{1, \dots, n\}, \text{ and } \quad (7)$$

$$\pi_{m,n}(q) = [\rho(n)]^{q-n} \pi_{m,n}(0) \prod_{k=1}^n \rho(k), \text{ for } q > n. \quad (8)$$

Let $W(m, n)$ refer to the expected delay a customer experiences (i.e., the expected time a customer waits before the trip begins) in a system with m vehicles and n drivers. Then, by virtue of Little's law, $W(m, n) =$

$(\sum_{q=0}^{\infty} q\pi_{m,n}(q)/\lambda) - s$, where s is the expected travel time between two uniformly drawn locations in the service region $\mathcal{C}$.

Before we proceed with the analysis of a system with remote drivers, let us first consider the benchmark case of a system with an equal number of drivers and vehicles (i.e., a system with $n = m$). When $n = m$, the stability condition in Lemma 1 can be restated per the following corollary.

Corollary 1. *A system with $n = m$ is stable if and only if $m > \gamma_2(\lambda)$, where $\gamma_2(\lambda)$ is defined in Proposition 1.*

Corollary 1 shows that, when $n = m$, the system is stable only in the supply-rich regime. The following proposition contrasts this result with the result for a system with remote drivers ($n \leq m$). Recall that q_1, q_2 , and q_3 are introduced in Proposition 1.

Proposition 3. *For a system with $n \leq m$, the following holds:*

- When $m < \gamma_1(\lambda)$, the system is unstable for any n .
- When $\gamma_1(\lambda) < m < \gamma_2(\lambda)$, the system is stable if and only if $q_1 < n < q_2$.
- When $m > \gamma_2(\lambda)$, the system is stable if and only if $n > q_3$.

Proposition 3 shows that, in the supply-limited regime, a system with remote drivers is never stable regardless of the number of drivers hired. A system in the supply-rich regime is stable if and only if the number of drivers is sufficiently high (above q_3). A system in the intermediate regime is stable if and only if the number of drivers is relatively moderate ($q_1 < n < q_2$). This result, perhaps surprisingly, indicates that a system that would otherwise be unstable (i.e., results in infinite delay) becomes stable (i.e., results in expected delay that is finite) by reducing the number of drivers, made

possible by the introduction of remote drivers. This is the case in the intermediate regime. The result can be explained again as follows. Reducing the number of drivers increases the number of idle vehicles, which shortens pickup times. When the supply of vehicles is relatively limited, this can lead to a net increase in service capacity. Obviously, when the number of vehicles is sufficiently small (the supply-limited regime), the benefit of reducing pickup times is not sufficient to overcome the lack of drivers, leading the system to be unstable.

Next, we provide a result analogous to the result in Theorems 1–3, showing that the impact of the number of drivers on expected delay can be nonmonotonic. Recall that q_1 , q_2 , and q_3 are introduced in Proposition 1. Define $\tilde{q}$ as the unique solution (between zero and m) to $\frac{d\rho(q)}{dq} = 0$.

Proposition 4. *The following holds:*

- i. When $\gamma_1(\lambda) < m < \gamma_2(\lambda)$, $W(m, n)$ decreases in n if $q_1 < n < \tilde{q}$ and increases in n if $\tilde{q} < n < q_2$.
- ii. When $m > \gamma_2(\lambda)$, $W(m, n)$ decreases in n if $q_3 < n < \tilde{q}$ and increases in n if $\tilde{q} < n \leq m$.

Proposition 4 shows that, in the intermediate and supply-rich regimes, customer delay may not be monotonic in the number of drivers, which implies that there is an optimal number of drivers ($\lfloor \tilde{q} \rfloor$ or $\lceil \tilde{q} \rceil$) under which the customer delay is minimized. This result can be explained again as follows. Reducing the number of drivers increases the number of idle vehicles, which shortens pickup times. This effect is particularly pronounced when there is a queue of customers waiting in the system. Therefore, when the vehicle supply is not severely constrained, it can lead to a net increase in service capacity.

Finally, we provide asymptotic results for the large market setting. Similar to the case with impatient customers, we carry out the analysis with $\mu(m, q)$ approximated by (4). For simplicity, we abuse notation and let $q^* = q_1$ when the system is in the intermediate regime and $q^* = q_3$ when the system is in the supply-rich regime. We show that, when the system is stable, the number of customers (or, equivalently, the number of busy vehicles) is concentrated around q^* .

Proposition 5. *Let $m_\lambda = \lfloor \alpha \lambda s \rfloor$ for $\alpha > 1$, $n_\lambda = \lfloor \beta m_\lambda \rfloor$ for $\beta \in (0, 1]$ such that the stability condition in Lemma 1 is satisfied. Then, we have $\lim_{\lambda \rightarrow \infty} |W(m_\lambda, n_\lambda) - (q^*/\lambda - s)| = 0$. Moreover, for any $\gamma \in (0, 1)$, we have $\lim_{\lambda \rightarrow \infty} \sum_{q \in ((1-\gamma)q^*, (1+\gamma)q^*)} \pi_{m, n}(q) = 1$.*

Because the number of customers is concentrated around q^* , we expect that, for maintaining a delay similar to the one in a system with equal numbers of drivers and vehicles, the number of drivers is not to be

significantly greater than q^* when the number of vehicles is large. We confirm this intuition and quantify the savings in the number of drivers in the following proposition.

Proposition 6. *Let $m_\lambda = \lfloor \alpha \lambda s \rfloor$ for $\alpha > 2$ and define $n_\lambda^* = \min\{n : |W(m_\lambda, n) - W(m_\lambda, m_\lambda)| \leq \epsilon\}$ for any $\epsilon > 0$. Then, $\limsup_{\lambda \rightarrow \infty} n_\lambda^*/m_\lambda \leq 1/\alpha$.*

Proposition 6 shows, analogously to the case with impatient customers, that the number of drivers can be reduced by at least a half, continuing to maintain a similar customer delay achieved by a system having as many drivers as vehicles.

4.1. Using the Nearest Dispatch Policy in Systems with Patient Customers

An alternative to the first dispatch policy we consider in the analysis of systems with patient customers is the so-called nearest dispatch policy (see, for example, Feng et al. 2021, Besbes et al. 2022, Wang et al. 2022). Under the nearest dispatch policy, instead of assigning a driver–vehicle pair to the customer who has been waiting the longest, a driver–vehicle pair is assigned to the customer who is nearest. Under such a policy, the dispatching policy is the same as the first dispatch policy when a customer arrives and there are multiple vehicles waiting. However, it is different when there are multiple customers waiting and a driver becomes available. Specifically, pickup times are now shorter the more there are customers waiting. In other words, the service rate $\mu(m, q)$ in this case is decreasing in the number of customers in system q when $q \leq n$ and is increasing in q when $q > n$. Put differently, the service rate is sensitive to the thickness of both the demand and supply sides.

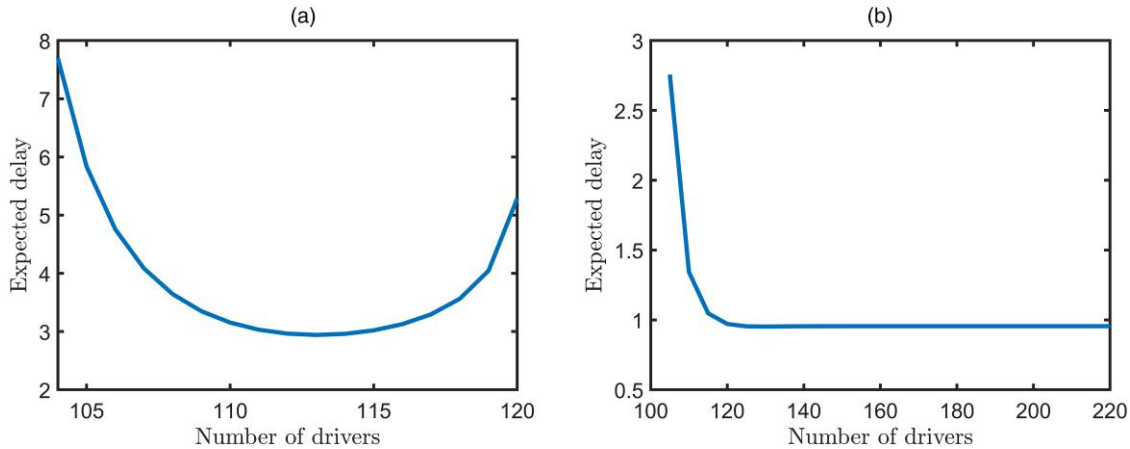
It is possible to extend our analysis of systems with patient customers to settings in which the nearest dispatch policy is used. In this case, the requirements placed on $\mu(m, q)$ are as follows: (1) there is a function $\bar{\mu}(m, q)$ that satisfies Assumptions 1 and 2 such that $\mu(m, q) = \bar{\mu}(m, q)$ when $q \leq n$, (2) $\mu(m, q)$ strictly increases in q when $q > n$, and (3) $\lim_{q \rightarrow \infty} \mu(m, q) = 1/s$ (i.e., pickup time approaches zero as the number of customers waiting in the queue approaches infinity).

Lemma 2 provides a condition for the system to be stable (the proof and additional discussion can be found in Online Appendix F.3).

Lemma 2. *Under the nearest dispatching policy, the system is stable if and only if $\lambda s/n < 1$.*

This lemma shows that the system is stable as long as $\frac{\lambda s}{n} < 1$. An implication of this (because $\frac{\lambda s}{n}$ is decreasing in n) is that it is no longer possible for reducing the number of drivers to stabilize a system that would otherwise be unstable. However, in a stable system, it is still possible

Figure 9. (Color online) The Impact of the Number of Drivers on Expected Customer Delay Under the Nearest Dispatching Policy



Notes. The results pertain to a system in which $\mu(m, q)$ is given by (4) if $q \leq n$ and $\mu(m, q) = 1/((s/\sqrt{(m-n+1)(q-n)}) + s)$ otherwise. The approximation is asymptotically exact (up to a geometry-dependent coefficient) when the origin–destination pair associated with each requested trip is uniformly distributed over a square service region. See theorem 5 in Wang et al. (2022) for the validation and more discussions. (a) Intermediate regime: $m = 120$, $\lambda = 10$, and $s = 10$. (b) Supply-rich regime: $m = 220$, $\lambda = 10$, and $s = 10$.

to reduce the delay experienced by customers by reducing the number of drivers (i.e., an analogous result to Proposition 4). This effect is illustrated for an example system in Figure 9. As shown in panel (a) of Figure 9, expected customer delay can be nonmonotone in the number of drivers; that is, $E[W(m, n)]$ is first decreasing and then increasing in n . Moreover, as shown in panel (b) of Figure 9, it continues to be possible, when the supply of vehicles is large, to reduce the number of drivers without significantly increasing delay. In Online Corollary F.4, we show that, in the limit, the reduction in the number of drivers is at least a half (i.e., an analogous result to Theorem 6).

4.2. Systems with Imperfectly Patient Customers

We conclude this section by noting that one may also consider the case in which customers are imperfectly patient. That is, customers are willing to wait but only up to a threshold (typically customer dependent) after which they abandon and leave without receiving service. Such a system is always stable if the distribution of the threshold has a finite mean. In Online Appendix E.6, we provide numerical results for a system in which customers who are not immediately matched with a vehicle and driver abandon after an exponentially distributed amount of time. As shown in Online Figure E.1, reducing the number of drivers can improve service level and reduce customer delay when the supply of vehicles is limited. When the supply of vehicles is ample, it is possible to significantly reduce the number of drivers without significantly affecting service level and customer delay.

5. Concluding Comments

In this paper, we explore the novel concept of tele-driving and examine the extent to which the number of drivers relative to the number of vehicles can be reduced. Among our findings, we show that, depending on the supply of vehicles relative to the workload, there is an opportunity for a system with remote drivers (appropriately sized) to significantly improve service level or significantly reduce the number of drivers without affecting service level. If customers are patient, we show that a system with remote drivers can stabilize an otherwise unstable system, reduce delay, or significantly reduce the number of drivers, maintaining a similar expected delay. Our analysis uncovers how the trade-off between shorter service times and more servers, brought about by remote driving, shapes these outcomes.

Additional discussion of aspects of our modeling, analysis, and results can be found in the Online Appendix, including a discussion of (a) the performance of systems with remote drivers versus systems with in-vehicle drivers, (b) systems in which a remote driver oversees multiple vehicles, and (c) mitigating the wild goose chase effect using alternative approaches such as not fulfilling demand when the number of idle vehicles falls below a certain threshold or imposing a matching radius in assigning vehicles to service requests (see Online Appendix I).

In this paper, we focus on a single driver of efficiency (the reduction in the number of drivers). As mentioned, there may be other benefits from tele-driving, including eliminating inefficiencies associated with drivers acting strategically (in settings in which the drivers can act

independently), increasing access to areas that are perceived by drivers as less desirable, and increasing demand by making the service more attractive to certain users (e.g., those who may feel uncomfortable riding with a stranger). There may also be benefits to drivers (e.g., by removing the requirement that drivers own a vehicle, increasing labor participation as drivers may work remotely at locations that are most convenient to them, and reducing income variability because of the reduction in the spatial mismatch between supply and demand). Quantifying these benefits, though each is likely to require a different approach, could provide interesting avenues for future research.

Acknowledgments

The authors thank the department editor, the associate editor, and three anonymous reviewers for valuable comments and suggestions.

Endnotes

¹ There are potentially other benefits to tele-operation: (1) it could reduce inefficiencies associated with drivers acting strategically by driving empty from locations perceived to be low demand to those perceived to be high demand (Ashkrof et al. 2023); (2) it can eliminate discriminatory behavior on the part of drivers toward travelers based on their race and gender (Ge et al. 2016, Tang et al. 2021); (3) it can increase the safety of both drivers and riders, particularly women who have been disproportionately the targets of in-vehicle assault and other criminal behavior (O'Brien 2022); (4) it removes the requirement that drivers own a vehicle (reducing their costs and the associated risks of owning a physical asset); and (5) it could broaden labor participation as drivers may work remotely at locations that are most convenient (Robotics Tomorrow 2022).

² See Online Appendix I for more discussions about the differences between tele-driving and traditional systems, including vehicle cost, labor cost, and the possibility that tele-driving is slower than in-vehicle driving, and their impacts on our main results.

³ We assume that vehicles can always find parking in a nearby location to where a customer is dropped off (e.g., at a metered parking space or a parking garage). We expect service providers to make arrangements with the cities in which they operate to allow for such parking, not unlike the arrangements free-floating car-sharing companies have made (see, for example, <https://eviecarshare.com/>). Alternatively, a service provider may invest in a network of designated hubs at which idle vehicles can be left. In this scenario, tele-drivers may spend more time parking the vehicle, which can be added as a third component to service time. These different assumptions on parking can be easily incorporated into the model, but do not affect our qualitative results.

⁴ The distribution of origin and destination locations can be general as long as Assumptions 1 and 2 hold. In Online Appendix H, we show that these assumptions hold for typical nonuniform traffic generated by large events, such as concerts and sporting events. In Section 3.4, to test the robustness of our findings, we provide numerical results for which the demand and service process are both data driven using real-world data.

⁵ In the case in which customers are impatient, this assumption can be relaxed. See Endnote 9 for more discussions.

⁶ Our analysis only requires that $\lim_{m \rightarrow \infty} \mu(m, q)$ exists, and the limit can depend on the distribution of origin and destination locations.

⁷ Our analysis only requires that $\mu(m, m)$ is nondecreasing in m .

⁸ In general, our model yields $\bar{m} < 10$ for a wide range of service region geometries and various approximations of the functional form of $\mu(m, q)$ that have been employed in the literature (see, for example, Feng et al. 2021, Kanoria 2021, Besbes et al. 2022).

⁹ Cheah and Smith (1994) show the stochastic equivalence between the $M/M(q)/1/l$ system and the $M/G(q)/1/l$ system. Therefore, all of our results for the system with impatient customers hold for general service time distributions.

¹⁰ Systems in which m and λ are small can be analyzed using the same approach though certain regimes identified in Proposition 1 may no longer exist. Given that small systems hold less significance, we choose to simplify the presentation and omit their analysis for brevity.

¹¹ Note that q_1 , q_2 , and q_3 depend on m and λ . For notational compactness, we do not express this dependence explicitly.

¹² We also simulate systems in which vehicles travel at a constant speed between locations using both the L_1 and L_2 distance metrics in service regions with different geometries. These simulations yield the same outcomes.

¹³ A similar phenomenon was observed empirically by Castillo et al. (2024) using data from the ride-hailing service Uber with pickup times and fulfilled demand varying significantly even when the number of vehicles stays the same.

¹⁴ Note that the bound does not depend on ϵ as $\lim_{\lambda \rightarrow \infty} [SL(m_\lambda, m_\lambda) - SL(m_\lambda, \lfloor (1 + \delta)q_3 \rfloor)] \rightarrow 0$ for any $\delta > 0$ when $\alpha > 2$. See the proof of Theorem 6 in Online Appendix D.3 for details.

¹⁵ See <https://www1.nyc.gov/site/tlc/about/data.page>.

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